

Object Oriented Programming Lab File



Submitted By:

Name: Hardik Lal

SAP ID: 500120516

Enrolment No: R2142230372

Batch: 4 (CCVT)

Experiment 1: Introduction to Java Environment

Answer The Following and submit it as a part of lab experiment work:

1. Explore and understand the role of JDK, JRE, and JVM
2. Install the latest available JDK and verify the Java Environment.
3. Create a Sample Hello World Program using a simple text editor (e.g., Notepad) and show the steps to compile and execute the program using the command prompt.
4. Display your name and complete address in different lines.
5. Designing a Visually Appealing Grade Sheet. Write a Java program to display a visually appealing grade sheet using escape sequences (\n, \t) and special characters (*).

Answer the following:

1. Explore and understand the role of JDK, JRE, and JVM

Ans:

These three components are essential for running and developing Java applications.

JVM (Java Virtual Machine)

- JVM is responsible for running Java programs.
- It converts Java bytecode (compiled code) into machine code that the operating system understands.
- It provides memory management, garbage collection, and security.
- JVM is platform-dependent but executes platform-independent bytecode.

JRE (Java Runtime Environment)

- JRE contains JVM and essential libraries to run Java applications.
- It does *not* include development tools like the compiler.
- If you only want to run Java programs (not develop them), installing JRE is enough.

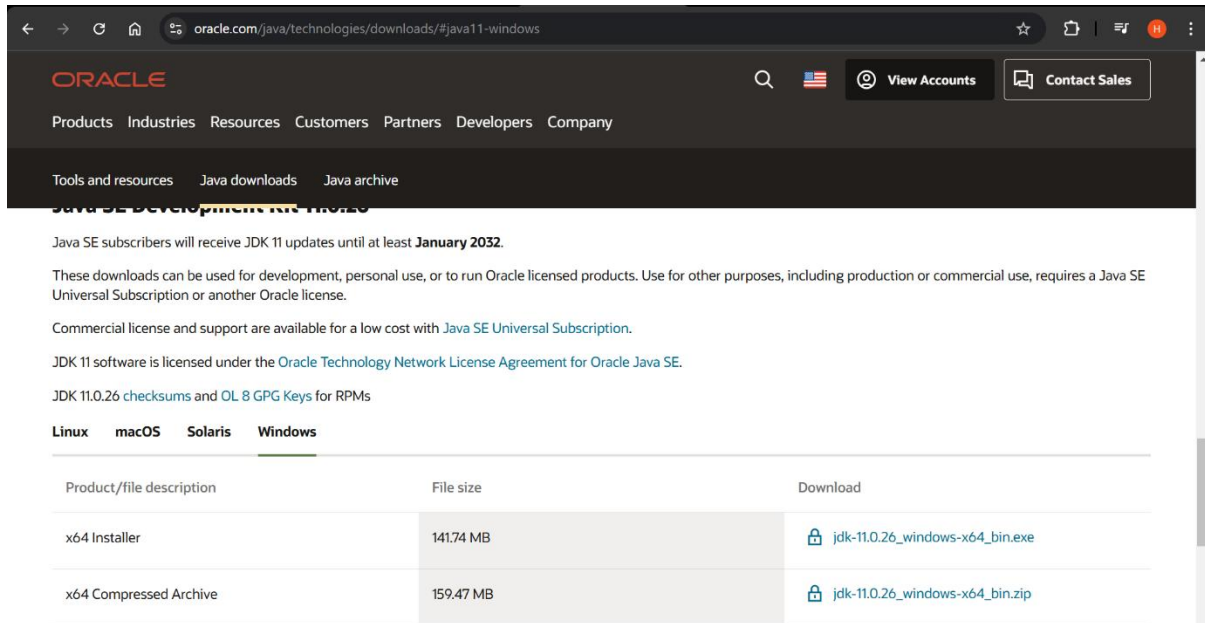
JDK (Java Development Kit)

- JDK includes JRE + additional tools for development (compiler, debugger, etc.).
- It is required for writing, compiling, and running Java programs.
- It contains javac (Java compiler) and java (Java runtime).

2. Install the latest available JDK and verify the Java Environment.

Ans:

Installation of latest version of JDK:



The screenshot shows the Oracle Java Downloads page for Windows. The page header includes the Oracle logo, a search icon, a US flag, and buttons for 'View Accounts' and 'Contact Sales'. The navigation menu lists 'Products', 'Industries', 'Resources', 'Customers', 'Partners', 'Developers', and 'Company'. Below the navigation menu, there are links for 'Tools and resources', 'Java downloads', and 'Java archive'. The main content area is titled 'Java SE Development Kit 11.0.26' and contains several paragraphs of text. The first paragraph states that Java SE subscribers will receive JDK 11 updates until at least January 2032. The second paragraph explains that these downloads can be used for development, personal use, or to run Oracle licensed products, but production or commercial use requires a Java SE Universal Subscription or another Oracle license. The third paragraph mentions that commercial license and support are available for a low cost with Java SE Universal Subscription. The fourth paragraph states that JDK 11 software is licensed under the Oracle Technology Network License Agreement for Oracle Java SE. The fifth paragraph provides links for JDK 11.0.26 checksums and OL 8 GPG Keys for RPMs. Below the text, there is a table with three columns: 'Product/file description', 'File size', and 'Download'. The table has two rows: 'x64 Installer' and 'x64 Compressed Archive'. The 'x64 Installer' row shows a file size of 141.74 MB and a download link for 'jdk-11.0.26_windows-x64_bin.exe'. The 'x64 Compressed Archive' row shows a file size of 159.47 MB and a download link for 'jdk-11.0.26_windows-x64_bin.zip'.

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Java SE Development Kit 11.0.26

Java SE subscribers will receive JDK 11 updates until at least **January 2032**.

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JDK 11.0.26 [checksums](#) and [OL 8 GPG Keys](#) for RPMs

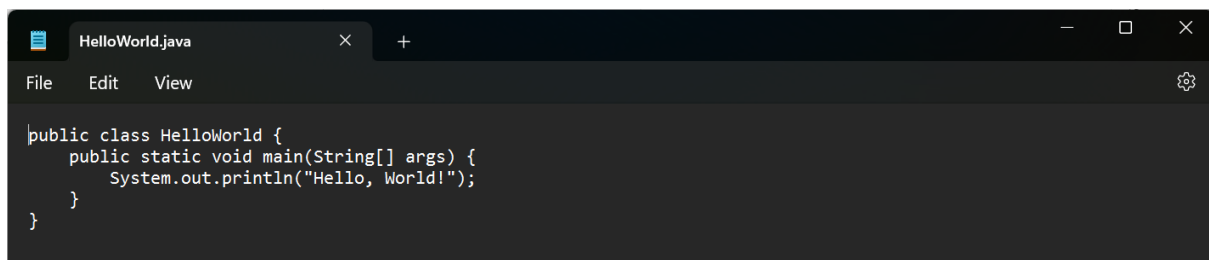
Linux **macOS** **Solaris** **Windows**

Product/file description	File size	Download
x64 Installer	141.74 MB	jdk-11.0.26_windows-x64_bin.exe
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3. Create a Sample Hello World Program using a simple text editor (e.g., Notepad) and show the steps to compile and execute the program using the command prompt.

Ans:

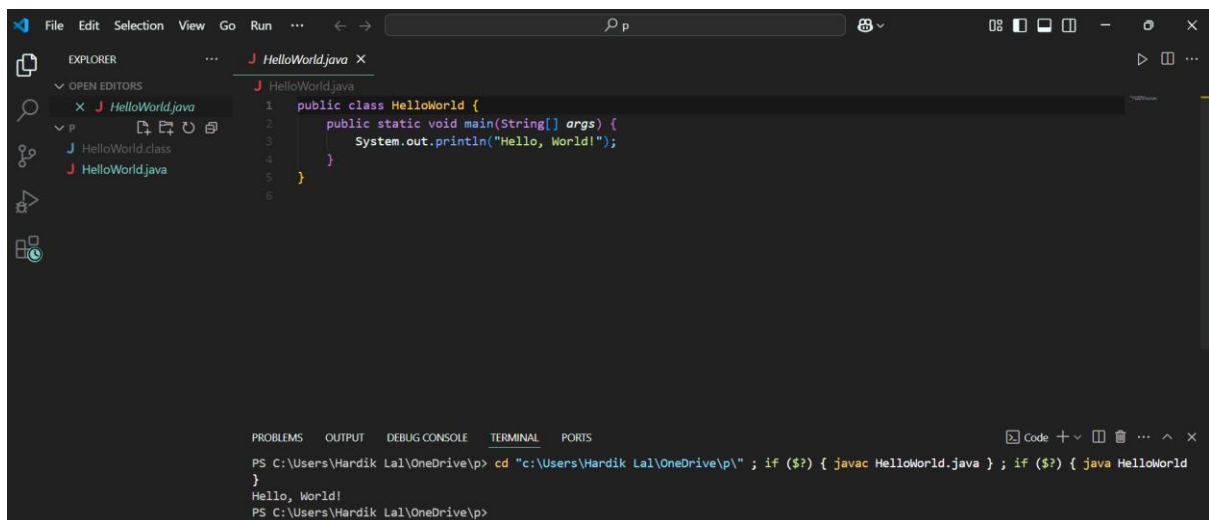
Write the Code in any text editor like notepad



A screenshot of a Notepad window titled 'HelloWorld.java'. The code inside is a simple Java program:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hello, World!");  
    }  
}
```

We can run the code in any compiler

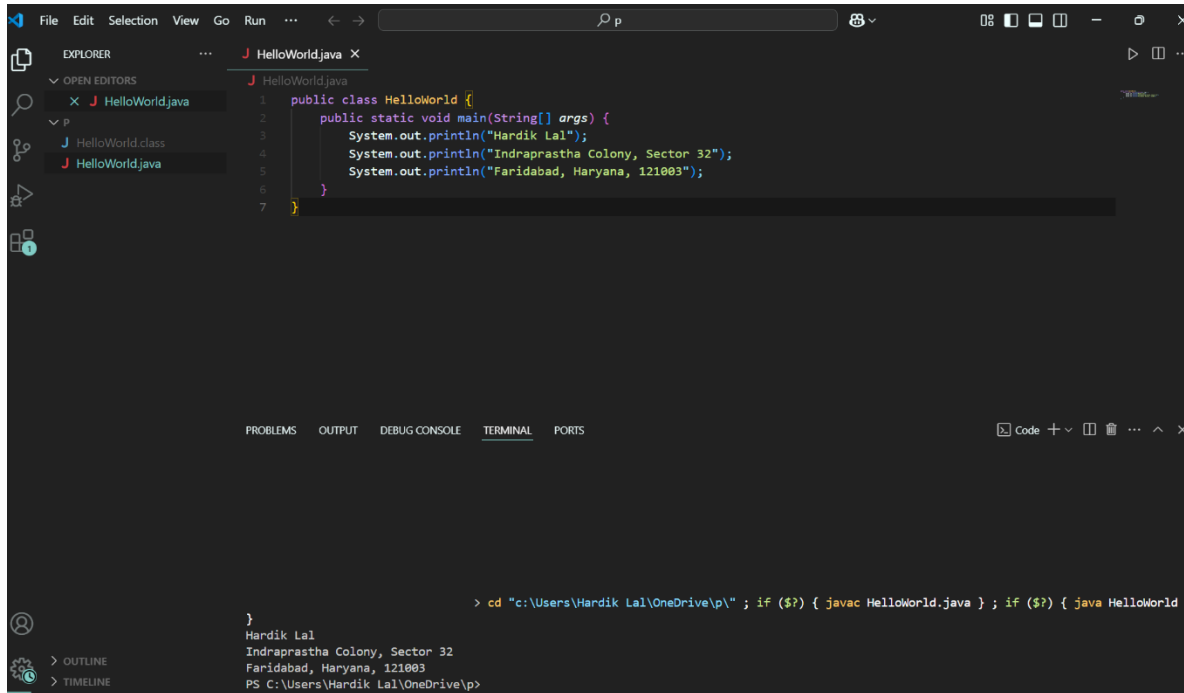


A screenshot of an IDE (likely VS Code) showing the same Java code as before. The terminal at the bottom shows the command prompt output:

```
PS C:\Users\Hardik Lal\OneDrive\p> cd "c:\Users\Hardik Lal\OneDrive\p\" ; if ($?) { javac HelloWorld.java } ; if ($?) { java HelloWorld }  
Hello, World!  
PS C:\Users\Hardik Lal\OneDrive\p>
```

4. Display your name and complete address in different lines.

Ans:



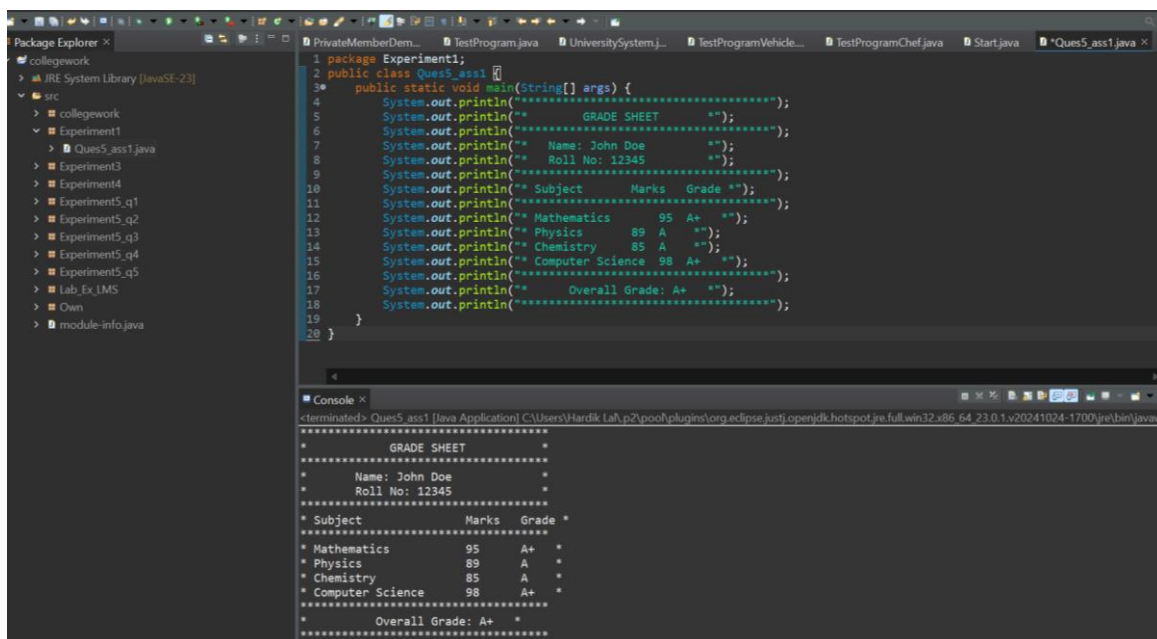
The screenshot shows an IDE with a file named `HelloWorld.java`. The code is as follows:

```
public class HelloWorld {  
    public static void main(String[] args) {  
        System.out.println("Hardik Lal");  
        System.out.println("Indraprastha Colony, Sector 32");  
        System.out.println("Faridabad, Haryana, 121003");  
    }  
}
```

The terminal output shows the execution of the program, displaying the name and address on three separate lines:

```
> cd "C:\Users\Hardik Lal\OneDrive\p"; if ($?) { javac HelloWorld.java }; if ($?) { java HelloWorld  
Hardik Lal  
Indraprastha Colony, Sector 32  
Faridabad, Haryana, 121003  
PS C:\Users\Hardik Lal\OneDrive\p>
```

5. Designing a Visually Appealing Grade Sheet. Write a Java program to display a visually appealing grade sheet using escape sequences (\n, \t) and special characters (*).



The screenshot shows an IDE with a file named `Ques5_ass1.java`. The code is as follows:

```
package Experiment1;  
public class Ques5_ass1 {  
    public static void main(String[] args) {  
        System.out.println("*****");  
        System.out.println("***** GRADE SHEET *****");  
        System.out.println("*****");  
        System.out.println("Name: John Doe");  
        System.out.println("Roll No: 12345");  
        System.out.println("*****");  
        System.out.println("Subject    Marks    Grade");  
        System.out.println("*****");  
        System.out.println("Mathematics    95    A+");  
        System.out.println("Physics        89    A");  
        System.out.println("Chemistry      85    A");  
        System.out.println("Computer Science 98    A+");  
        System.out.println("*****");  
        System.out.println("Overall Grade: A+");  
        System.out.println("*****");  
    }  
}
```

The console output shows the execution of the program, displaying a visually appealing grade sheet:

```
<terminated> Ques5_ass1 [Java Application] C:\Users\Hardik Lal\p2\poo\plugins\org.eclipse.justi.openjdk.hotspot.jre.full.win32.x86_64.23.0.1.v20241024-1700\jre\bin\javaw  
*****  
***** GRADE SHEET *****  
*****  
Name: John Doe  
Roll No: 12345  
*****  
Subject    Marks    Grade  
*****  
Mathematics    95    A+  
Physics        89    A  
Chemistry      85    A  
Computer Science 98    A+  
*****  
Overall Grade: A+  
*****
```

